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The relationship between multiple concussions and multidimensional sleep quality in collegiate-aged, active athletes

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ABSTRACT

Objectives: Determine the association of cumulative concussion and repetitive head impacts with self-reported sleep quality in healthy collegiate-aged athletes.
Methods: Collegiate-aged athletes (N = 212; mean age 21.00, 62.7% male) completed semistructured interviews for sport and concussion history and the Pittsburgh Sleep Quality Index (PSQI). Number of concussions was retrospectively determined based on the 1993 American Congress of Rehabilitation Medicine (ACRM) criteria; repetitive head impact was measured based on the cumulative years of contact sport exposure. Associations of number of concussions and repetitive head impact exposure with global PSQI score, overall poor (PSQI > 5) vs. good sleep, and binarized subscale scores were tested. Secondary analyses were conducted using alternative concussion criteria and metrics of repetitive head impact.
Results: The number of prior concussions was associated with higher PSQI global scores (B(SE) = 0.50(0.13), $p < .001$). Participants with more concussions were more likely to be poor sleepers (OR = 1.52, $p < .001$), report poorer sleep quality (OR = 1.29, $p = .037$), longer sleep latency (OR = 1.34, $p = .005$), more sleep disturbances (OR = 1.56, $p = .001$), increased use of sleep medications or sleep aids (OR = 1.35, $p = .008$), and more sleep-related daily dysfunction (OR = 1.38, $p = .002$). Similar results were observed for alternative definitions of concussion. No metric of repetitive head impact was associated with any sleep quality metric.
Conclusions: More prior concussions, but not repetitive head impact exposure, are associated with worse self-reported sleep, with subscale analyses showing concussion history associated with multiple aspects of subjective sleep quality rather than sleep quantity. Sleep represents an important factor to consider for future research aimed at characterizing and ultimately preventing adverse long-term health outcomes associated with concussion history.
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Introduction

Over the last decade, there has been rising concern over the potential cumulative and long-lasting effects of prior concussions, also known as mild traumatic brain injuries (mTBI), and repetitive head impact (RHI) exposure on neurological health among contact sport athletes. Sleep, including sleep quality, is inherently intertwined with brain health, with poor sleep representing both a risk factor for and consequence of a variety of neurological and psychiatric conditions that have been associated with repeat concussion and RHI.[1–4] However, despite the important health implications of dysregulated sleep, potential associations of sleep quality with RHI and cumulative concussions remain relatively unexplored (e.g., in contrast to cognition or mood-related outcomes).

Sleep-related problems such as sleep disturbance or chronic sleep disorders are a common consequence of traumatic brain

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injuries (TBI) of all severity (i.e., mTBI and moderate-to-severe TBI), observed both acutely following TBI and even years postinjury.[5–9] Acute sleep abnormalities are also well-documented following concussion, particularly in the context of persistent symptom burden.[10] Postconcussion sleep disturbances and poor sleep quality are also associated with prolonged injury recovery,[11] worse concussion-related symptoms,[12,13] and even abnormalities in white matter microstructure.[14] Moreover, greater acute sleep disturbances are also observed in patients with repeat concussion in a dose-dependent manner.[15]

As previously mentioned, however, fewer studies have investigated the chronic or cumulative effects of concussion or RHI on sleep. In one such study, concussion history was predictive of sleep quality in former collegiate American football players[16]; in another, there was an association between younger age at first exposure (AFE) and sleep disturbances in former NFL players who had at least 1 prior concussion.[17] RHI was identified as a potential risk factor for developing rapid eye movement (REM) sleep disorders within former American football players diagnosed with chronic traumatic encephalopathy (CTE) postmortem.[18] Most relevant to the current study, a study in student-athletes found that those with 3 or more concussions reported more sleep disturbances than athletes with no prior concussion.[19]

Given the relative dearth of studies investigating this topic, many questions remain. First, much of the prior work has focused on retired athletes, typically male former American football players; less work has examined these associations in younger, active male and female athletes across a variety of sports. In addition, the limited prior evidence linking poor sleep with RHI and/or concussion has used various alternative methods to retrospectively quantify concussion and RHI history. This could have important implications, as findings could conceivably depend on the diagnostic criteria used by the study to operationalize concussion (e.g., symptom-based diagnosis vs. the need for alterations in mental status),[20] as well as the specific metric of RHI obtained (e.g., total years of contact sport participation and the age of first exposure to contact sport).[21–23] Finally, most prior studies assessing the association of concussion or RHI history with subjective measures of sleep have used a global metric of sleep quality. Overall sleep quality, however, can be impacted or driven by a variety of factors, such as exposure to disturbances, inability to fall asleep in a timely manner, and short durations of sleep. It is conceivable that concussion history and RHI are associated with specific subcomponents of sleep; however, to our knowledge, this has not been previously investigated. Thus, the full extent to which concussion and RHI are associated with sleep problems is still unknown.

The goal of this work was to assess the association of lifelong concussion history and RHI exposure with subjective reporting of overall sleep quality in a cohort of active, healthy collegiate-aged athletes. It was hypothesized that a greater history of concussion and RHI would independently be associated with poorer subjective sleep quality. The second goal was to explore the extent to which individual aspects of sleep are associated with prior concussion and RHI exposure. Finally, as previously stated, there is currently no consensus method for characterizing RHI exposure (e.g., using total years of exposure vs. age of first exposure), and even the retrospective diagnosis of concussion relies upon the operational definition used. Therefore, an exploratory goal was to compare the association of sleep quality with varying definitions of concussion and RHI metrics.

Participants and methods

Participants were enrolled between December 2018 and January 2023 as part of a project aimed at assessing the chronic effects of concussion and contact sport exposure in collegiate-aged athletes (Project ChronEx). Participants were eligible for the study if aged 18-26 years, actively involved in a sport within 12 months, and proficient in English. Exclusion criteria included self-reported psychiatric, neurodevelopmental (including attention-deficit/hyperactivity disorder), neuroendocrine, severe neurological, or severe autoimmune diseases; use of neuroactive medication; history of moderate or severe TBI; concussion within the past 6 months or incomplete recovery from previous concussion; excessive alcohol or substance use (defined as scores ≥2 on the CAGE Questions Adapted to Include Drug Use, CAGE-AID); contra-indication to study procedures; and injury/illness precluding participation in study protocol. This study was approved by the Institutional Review Board at the Medical College of Wisconsin; all participants gave written informed consent.

Interviews for prior concussion and RHI exposure

A detailed history of sport exposure and head injury across the lifespan was collected from semistructured interviews, as previously described.[22,24] The primary criteria for retrospectively diagnosing concussions for the current study was broadly based on criteria consistent with the 1993 definition of mTBI provided by the American Congress of Rehabilitation Medicine (herein referred to as the ACRM definition).[25] Specifically, injuries were included in the final count if they were accompanied by loss of consciousness (≤30 minutes; LOC); post-traumatic amnesia (≤24 hours; PTA); retrograde amnesia; or alteration in mental status (e.g., being dazed or confused). The number of prior concussions based on criteria consistent with the 2016 International Conference on Concussion in Sport Group definition (CISG definition)[26] was also obtained as a secondary metric of concussion history. In this definition, any prior injury with LOC (≤30 minutes), PTA (≤24 hours), retrograde amnesia, alteration in mental status (i.e., being dazed or confused), or at least two symptoms from the SCAT-5 symptom checklist was included in the final count. The main difference in this definition compared to the ACRM definition is that the presence of symptoms alone without alteration in mental status or consciousness is sufficient for diagnosis. Finally, because prior work has demonstrated that concussions associated with LOC or PTA result in greater acute physiological injury (e.g., greater acute elevation of blood biomarkers and abnormalities in brain structure)[27,28] and increased risk of postinjury sleep disturbances,[29] the total number of concussions with LOC and/or PTA (concussions with LOC/PTA) was also determined for each participant.

To quantify cumulative exposure to RHI, detailed information regarding participation in contact/collision sports (e.g., American football, rugby, wrestling, soccer, men's lacrosse, men's ice hockey, and combat sports) was collected for each participant across the lifespan (i.e., youth through collegiate sports). For the current study, the cumulative number of years of exposure across all contact/collision sports (years of exposure) serves as the primary metric of RHI. In addition, two additional metrics of RHI were calculated as previously described.[22] First, AFE to contact sport has been linked with adverse outcomes in former athletes, with AFE occurring before 12 years of age being identified as a potential criteria age cutoff.[21,30,31] Therefore, for the current study, AFE was categorized as occurring before 12 years of age (< 12) or either occurring after 12 or having no contact sport experience (AFE 12+/none). Finally, exposure criteria have been recently proposed for traumatic encephalopathy syndrome (TES), the clinical syndrome proposed to be associated with chronic traumatic encephalopathy, as follows: "extensive" being > 11 years of contact sport exposure including ≥2 years at high school or later levels; "substantial" being ≥5 years of contact sport exposure including ≥2 years at high school or later levels; and "nonsubstantial."[23] In the current study, a binary TES variable was derived from the cumulative years of exposure for each participant based on distribution of groups ("extensive" vs. "nonextensive").

Subjective sleep assessment

Subjective sleep quality was assessed using the Pittsburgh Sleep Quality Index (PSQI). The PSQI is composed of seven subcomponents

of sleep that together create a global composite sleep score. The seven subcomponents are subjective sleep quality; latency (time in minutes it takes to fall asleep); sleep duration (total hours spent asleep); sleep efficiency (ratio of hours spent in bed vs. asleep); sleep disturbances; use of sleep medication or sleep aids to fall asleep; and sleep-related daily dysfunction. A score from zero to three is computed for each subscale, with higher scores indicating worse sleep/more sleep difficulties. These 7 subscores are totaled to create a global score from zero to 21, with higher scores again indicating worse sleep/more sleep difficulties. A cutoff score of more than 5 on the global score was originally proposed as the cutoff to identify significantly poorer sleep, with a sensitivity of 89.6% and specificity of 86.5% in validation studies.[32]

Statistical analysis

Data analyses were run using IBM SPSS Statistics Software Version 26. General linear models tested associations of number of prior concussions and cumulative RHI with global PSQI scores. The primary model included the number of concussions based on the ACRM definition and the years of exposure as the RHI metric. A secondary binary logistic regression was conducted to determine if the number of prior concussions and years of exposure were associated with poor sleep based on a PSQI cutoff > 5. Similarly, separate logistical regression models were fit to investigate the association of the number of prior concussions and years of exposure with individual subscales of the PSQI (i.e., quality, latency, duration, efficiency, disturbances, use of sleep medications, and daily dysfunction). Subscale scores were positively skewed and zero-inflated; thus, subscales were binarized based on the distribution of responses for analyses (see Supplementary Materials for distributions and cut-offs used for subscales). Sex was included as a covariate for all models.

Similar models were used to examine the association of PSQI scores with prior concussion based on varying diagnostic criteria. Two alternative sets of models were fit for prior concussion as above using (1) the number of concussions based on CISG definition and (2) the number of prior concussions with either LOC or PTA. For each of these models, years of contact sport exposure and sex were included as covariates. The association of alternative metrics of RHI (the categorial AFE and TES variables defined above, as well as current sport categorized as contact sport (CS) or noncontact sport (NCS)) were assessed using similar models as above, with prior concussion (ACRM definition) and sex included as covariates. A nominal *p*-value of .05 (two-tailed) was used to determine significance for all models.

Finally, sensitivity analyses were conducted to determine if (a) exclusion of current noncontact sport athletes, (b) exclusion of athletes with no prior concussion, or (c) the inclusion of additional potential confounding variables in the statistical models impacted observed results. As reported and discussed in the Supplement, these sensitivity analyses showed only minor differences from the main analyses reported in the manuscript.

Results

A total of 212 collegiate-aged athletes are included in this study (see Fig. 1). The average participant age was 21.00 years (SD = 1.69), and the sample was 62.7% male. Most participants reported their race as white (89%), and their ethnicity as not Hispanic or Latino (92%). Represented current sports included American football, soccer, rugby, lacrosse, ice hockey, softball, baseball, basketball, volleyball, track and field, cross country, golf, tennis, cheerleading, water polo, swim/dive, and wrestling. See Table 1 for complete demographics.

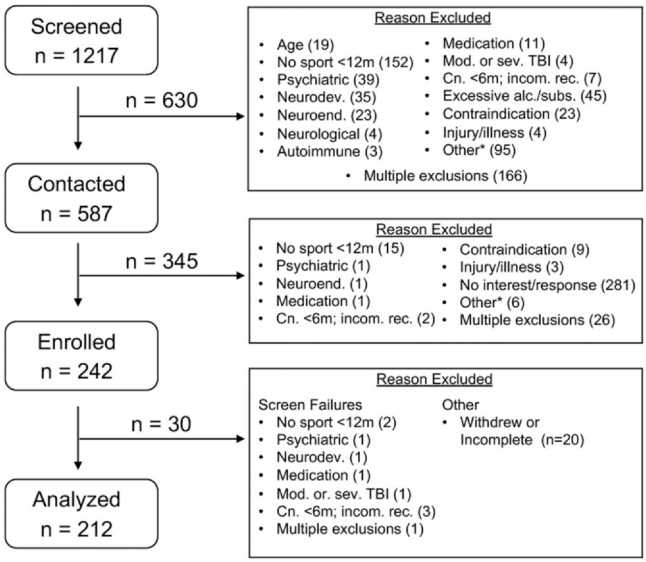

Fig. 1. Enrollment flow chart. Consort diagram for the study. Participants could have multiple reasons for exclusion at each step. Other* includes individuals not contacted by staff in error or to avoid cohort imbalance. Screen failures were due to disclosure of exclusionary factors during the study visit that were not disclosed initially. Alc./subs., alcohol/substance use; Cn. < 6 m; incom. Rec., concussion within the last 6 months or incomplete recovery from prior concussion; Neurodev., neurodevelopment; neuroend., neuroendocrine, mod. or sev; TBI, moderate or severe traumatic brain injury; < 12 m, less than 12 months

Table 1
Study demographics

Demographic variable	Full sample	
	N	%
Number of participants	212	–
Age M(SD)	21.00(1.69)	–
Sex		
Female	79	37.3
Male	133	62.7
Race		
White	189	89.2
Black or African-American	17	8.0
Native Hawaiian or Other Pacific Islander	1	0.5
Asian	1	0.5
Other	4	1.9
Ethnicity		
Not Hispanic or Latino	195	92.0
Hispanic or Latino	16	7.5
Unknown	1	0.5
Number of concussion, median [min, max]		
ACRM Criteria	1[0,8]	–
CISG Criteria	1[0,9]	–
Post-traumatic amnesia or loss of consciousness	0[0,4]	–
Age of first exposure		
Younger than 12	145	68.4
None or 12 or older	67	31.6
Traumatic encephalopathy syndrome criteria		
Extensive	117	55.2
Not extensive	95	44.8
Current sport[a]		
Contact	139	65.6
Noncontact	73	34.4
Pittsburg Sleep Quality Index Total Score M(SD)	4.58 (2.75)	–
Pittsburg Sleep Quality Index Scores > 5		
Yes	61	28.8
No	151	71.2

Abbreviations: ACRM, American Congress of Rehabilitation Medicine; CISG, Concussion in Sport Group; M, mean; SD, standard deviation.

[a] For the current study, contact sports included American football, soccer, rugby, wrestling, men's lacrosse, and men's ice hockey.

Table 2
Association between number of concussions and Pittsburgh Sleep Quality Index global score

	General linear model[a]				Logistic regression model[b]				
	B	SE	t	*p*	B	SE	*p*	OR	95% CI
Primary model (ACRM)									
Number of concussions	0.50	0.13	4.00	**< .001**	0.42	0.11	**< .001**	1.52	1.23,1.88
Contact sport exposure	0.03	0.03	1.02	.31	− 0.01	0.02	.78	0.99	0.95,1.04
Secondary model (CISG)									
Number of concussions	0.44	0.11	4.04	**< .001**	0.39	0.10	**< .001**	1.48	1.23,1.79
Contact sport exposure	0.03	0.03	0.99	.33	− 0.01	0.02	.75	0.99	0.95,1.04
Secondary model (PTA/LOC)									
Number of concussions	0.71	0.24	2.96	**.003**	0.48	0.19	**.011**	1.62	1.12,2.36
Contact sport exposure	0.03	0.03	1.24	.22	0.00	0.02	.98	1.00	0.96,1.05

Abbreviations: ACRM, American Congress of Rehabilitation Medicine; CI, confidence interval; CISG, Concussion in Sport Group; LOC, loss of consciousness; OR, odds ratio; PTA, post-traumatic amnesia; SE, standard error.
p-values < .05 are bolded.
[a] Global PSQI score is a continuous outcome variable.
[b] Global PSQI score is a binarized outcome variable using a cut-off score of more than 5 to identify poor vs. good sleepers.

Primary model: ACRM definition of concussion and years of exposure

A total of 105 participants had no history of concussion based on the ACRM definition; the remaining 107 participants had a history of concussion ranging from 1 to 8 concussions. Consistent with our hypothesis, a greater number of prior concussions was significantly associated with higher global PSQI scores, $p = < .001$, standardized beta = 0.27 (see Table 2 for associated statistics). The association of years of exposure with global PSQI scores was not significant. Similarly, the logistic regression model showed that a greater number of prior concussions was associated with increased risk of being a poor sleeper (PSQI global score of more than 5), $p < .001$. For each additional concussion, there was a 52% increase in the odds for experiencing poor sleep (odds ratio [OR] = 1.52). Again, there was no association between years of exposure and sleep (Table 2).

A greater number of prior concussions was also significantly associated with multiple PSQI subcomponents (see Table 3 for all associated statistics), including poorer sleep quality ($p = .037$, OR = 1.29); longer sleep latency ($p = .005$, OR = 1.34); more sleep disturbances ($p = .001$, OR = 1.56); increased use of sleep medications or sleep aids ($p = .008$, OR = 1.35); and more sleep-related daily dysfunction ($p = .002$, OR = 1.38). The number of prior concussions was not significantly associated with sleep duration or sleep efficiency (*ps* > .05). There were no effects of years of exposure (*ps* > .05).

Table 3
Association between concussion and Pittsburgh Sleep Quality Index subscales

	Number of concussions primary (ACRM)					Contact sport exposure				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	0.25	0.12	**.037**	1.29	1.02,1.63	0.03	0.03	.35	1.03	0.97,1.09
Latency	0.30	0.10	**.005**	1.34	1.10,1.65	0.00	0.02	.89	1.00	0.95,1.04
Duration	0.11	0.10	.27	1.12	0.92,1.37	0.02	0.02	.36	1.02	0.98,1.07
Efficiency	0.03	0.10	.76	1.03	0.85,1.25	0.02	0.02	.44	1.02	0.98,1.06
Disturbances	0.45	0.14	**.001**	1.56	1.19,2.05	0.00	0.04	.94	1.00	0.93,1.07
Medication	0.30	0.11	**.008**	1.35	1.08,1.68	0.01	0.03	.82	1.01	0.96,1.06
Daily Dys.	0.32	0.10	**.002**	1.38	1.13,1.69	0.00	0.02	.90	1.00	0.96,1.04
	Number of concussions secondary (CISG)					Contact sport exposure				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	0.25	0.11	**.019**	1.29	1.04,1.59	0.03	0.03	.36	1.03	0.97,1.09
Latency	0.23	0.09	**.012**	1.25	1.05,1.50	0.00	0.02	.92	1.00	0.95,1.04
Duration	0.12	0.09	.19	1.12	0.94,1.34	0.02	0.02	.37	1.02	0.98,1.07
Efficiency	0.09	0.09	.29	1.10	0.93,1.30	0.01	0.02	.50	1.02	0.97,1.06
Disturbances	0.31	0.12	**.011**	1.36	1.07,1.73	0.00	0.04	.96	1.00	0.93,1.08
Medication	0.31	0.10	**.002**	1.36	1.12,1.65	0.01	0.03	.86	1.01	0.95,1.06
Daily Dys.	0.21	0.09	**.017**	1.23	1.04,1.45	0.00	0.02	.96	1.00	0.96,1.04
	Number of concussions secondary (LOC/PTA)					Contact sport exposure				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	0.40	0.22	.07	1.49	0.97,2.29	0.03	0.03	.30	1.03	0.97,1.09
Latency	0.29	0.19	.13	1.33	0.92,1.94	0.00	0.02	.92	1.00	0.96,1.05
Duration	0.19	0.19	.32	1.21	0.83,1.77	0.02	0.02	.33	1.02	0.98,1.07
Efficiency	0.04	0.19	.82	1.04	0.72,1.51	0.02	0.02	.43	1.02	0.98,1.06
Disturbances	0.55	0.25	**.025**	1.73	1.07,2.81	0.01	0.04	.87	1.01	0.94,1.08
Medication	0.38	0.20	.06	1.46	0.98,2.18	0.01	0.03	.69	1.01	0.96,1.06
Daily Dys.	0.38	0.19	**.043**	1.46	1.01,2.12	0.00	0.02	.92	1.00	0.96,1.04

Abbreviations: ACRM, American Congress of Rehabilitation Medicine; CI, confidence interval; CISG, Concussion in Sport Group; Daily Dys., daily dysfunction; LOC/PTA, loss of consciousness or post-traumatic amnesia; OR, odds ratio; PSQI, Pittsburgh Sleep Quality Index; SE, standard error.
p-values < .05 are bolded.

Alternative definitions of concussion

A total of 87 participants had no history of concussion based on CISG criteria; the remaining 125 participants had a history of concussion ranging from 1 to 9 concussions. As in the primary model, a greater number of prior concussions was significantly associated with higher global PSQI scores, $p = < .001$, standardized beta = 0.27 (see Table 2 for associated statistics). Similarly, logistic regression showed that a greater number of prior concussions was associated with increased risk of being a poor sleeper (PSQI score of more than 5), $p < .001$, OR = 1.48. Further investigation of subcomponents (see Table 3) showed that a greater number of prior concussions based on CISG criteria was significantly associated with poorer sleep quality ($p = .019$, OR = 1.29); longer sleep latency ($p = .012$, OR = 1.25); more sleep disturbances ($p = .011$, OR = 1.36); increased use of sleep medications or aids ($p = .002$, OR = 1.36); and more sleep-related daily dysfunction ($p = .017$, OR = 1.23). As in the primary model, the number of prior concussions was not significantly associated with sleep duration or sleep efficiency ($ps > .05$).

A total of 160 participants had no history of concussion with either LOC or PTA; the remaining 52 participants reported 1 to 4 concussions with either LOC or PTA. As in the primary model, a greater number of prior concussions with PTA or LOC was significantly associated with higher global PSQI scores, $p = .003$, standardized beta = 0.20 (see Table 2). Similarly, logistic regression showed that a greater number of prior concussions with PTA or LOC was associated with increased risk of being a poor sleeper (PSQI score of more than 5), $p = .011$, OR = 1.62. Further investigation of subcomponents (see Table 3) showed that a greater number of prior concussions with PTA or LOC was significantly associated with greater sleep disturbances ($p = .025$, OR = 1.73) and more daily dysfunction ($p = .043$, OR = 1.46).

Alternative metrics of RHI

To further explore possible associations with RHI exposure, additional analyses were conducted using AFE and extent of exposure delineated using recently proposed TES criteria as the proxy for RHI. A total of 145 athletes reported an AFE younger than the age of 12 and the remaining 67 athletes reported an AFE equal to or greater than the age of 12, or no AFE history (noncontact athletes). Based on TES criteria, 117 athletes were found to have extensive exposure and the remaining 95 athletes were found to have nonsubstantial or not extensive exposure. Finally, 139 athletes were contact sport athletes and 73 were noncontact sport athletes. AFE, TES, and current sport (CS vs. NCS) were not significantly associated with global sleep score or any subcomponents (see Tables 4 and 5).

Discussion

Previous literature has identified increased prevalence of sleep disorders and complaints as a common consequence of TBI,[5-9] but the cumulative impact of repeat concussion and RHI on subjective sleep quality in otherwise young, healthy, and active athletes has not been definitively established. In the current study, a greater number of prior concussions in young-adult athletes was significantly associated with worse self-reported overall sleep quality. Analysis of subcomponents of the PSQI found that athletes with a greater history of concussion reported worse sleep quality, greater sleep latency, more sleep disturbances or waking, increased use of sleep aids or medication, and increased reports of daily dysfunction. Prior concussion was not associated with actual sleep duration or sleep efficiency. Similar results were observed when using varying definitions for retrospectively identifying prior concussion, though effect sizes were qualitatively larger for diagnostic criteria based on the presence of more severe acute injury characteristics (e.g., alterations in consciousness vs. symptoms alone). In contrast, RHI exposure was not associated with sleep quality, regardless of the metric used (i.e., cumulative years, AFE, TES criteria). Overall, results demonstrate that cumulative history of concussion, not RHI exposure, affects sleep quality in a multifaceted manner, even among younger, active athletes.

Current results demonstrating an association of concussion history with worse self-reported sleep quality in active (i.e., not former), non-recently concussed athletes are consistent with the only prior study, to our knowledge, to directly address this issue. Specifically, Blake and colleagues assessed self-reported sleep disturbances (i.e., fatigue, sleeping too much or too little, restlessness, difficulty falling or staying asleep) and found that athletes with 3 or more concussions reported significantly more disturbances than those with no prior concussion.[19] Our results demonstrate a potential dose-dependent effect of the number of prior concussions with sleep, an effect that was observed both when treating sleep as a continuous measure and when comparing poor vs. good sleepers based on recommended cut-offs. Sleep quality is a fundamental component of overall health, and poor sleep quality is associated with a variety of adverse health outcomes, including those that have been previously associated with TBI, such as major depressive disorder and neurodegenerative disease.[1,4] Directly related to concussion, poor sleep (e.g., insomnia and daytime sleepiness) has been proposed to be a significant and underappreciated indicator of future concussion risk.[33] Together, these results suggest a bidirectional relationship between concussion and sleep where prior concussion is associated with poor sleep which in turn may predispose individuals to suffering additional concussions. It is worth noting, however, that given the cross-sectional nature of the current study,

Table 4
Association between additional repetitive head impact metrics and Pittsburgh Sleep Quality Index global score

	General linear model[a]				Binary logistic regression model[b]				
	B	SE	t	*p*	B	SE	*p*	OR	95% CI
Traumatic encephalopathy syndrome (TES) model									
TES	0.17	0.40	0.44	.66	− 0.03	0.34	.94	0.98	0.50,1.90
Number of concussions	0.51	0.12	4.09	**< .001**	0.41	0.11	**< .001**	1.51	1.23,1.87
Age of first exposure (AFE) model									
AFE	0.16	0.41	0.38	.71	0.06	0.36	.87	1.06	0.53,2.14
Number of concussions	0.51	0.12	4.12	**< .001**	0.42	0.11	**< .001**	1.51	1.23,1.87
Contact vs. Noncontact sport (CS/NCS) model									
CS/NCS	0.35	0.41	0.85	.40	− 0.06	0.36	.86	0.94	0.47,1.89
Number of concussions	0.50	0.12	4.04	**< .001**	0.41	0.11	**< .001**	1.51	1.22,1.86

Abbreviations: CI, confidence interval; CS,contact sport; OR, odds ratio; NCS, noncontact sport; SE, standard error.
p-values < .05 are bolded.
[a] Global PSQI score is a continuous outcome variable.
[b] Global PSQI score is a binarized outcome variable using a cut-off score of more than 5 to identify poor vs. good sleepers.

Table 5
Association between additional repetitive head impact metrics and Pittsburgh Sleep Quality Index subscales

	Traumatic encephalopathy syndrome criteria					Number of concussions				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	− 0.07	0.45	.88	0.93	0.38,2.27	0.27	0.12	**.028**	1.30	1.03,1.65
Latency	− 0.07	0.34	.84	0.93	0.48,1.83	0.29	0.10	**.005**	1.34	1.09,1.64
Duration	− 0.07	0.34	.85	0.94	0.48,1.81	0.12	0.10	.23	1.13	0.93,1.38
Efficiency	− 0.06	0.32	.85	0.94	0.51,1.76	0.04	0.10	.70	1.04	0.86,1.26
Disturbances	0.25	0.54	.64	1.29	0.44,3.74	0.45	0.14	**.001**	1.57	1.20,2.06
Medication	− 0.08	0.40	.85	0.93	0.42,2.04	0.30	0.11	**.007**	1.35	1.08,1.68
Daily Dys.	− 0.07	0.31	.81	0.93	0.51,1.70	0.32	0.10	**.002**	1.38	1.12,1.68
	Age of first exposure					Number of concussions				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	− 0.25	0.50	.61	0.78	0.29,2.07	0.27	0.12	**.028**	1.31	1.03,1.65
Latency	− 0.16	0.36	.67	0.86	0.42,1.75	0.29	0.10	**.005**	1.34	1.09,1.64
Duration	− 0.03	0.36	.94	0.97	0.49,1.95	0.13	0.10	.22	1.13	0.93,1.38
Efficiency	0.13	0.33	.69	1.14	0.59,2.20	0.04	0.10	.68	1.04	0.86,1.27
Disturbances	− 0.29	0.58	.62	0.75	0.24,2.33	0.45	0.14	**.001**	1.56	1.19,2.04
Medication	− 0.20	0.44	.66	0.82	0.35,1.94	0.30	0.11	**.007**	1.35	1.09,1.68
Daily Dys.	− 0.12	0.33	.71	0.89	0.47,1.68	0.32	0.10	**.002**	1.38	1.12,1.68
	Contact vs. Noncontact sport					Number of concussions				
	B	SE	*p*	OR	95% CI	B	SE	*p*	OR	95% CI
Quality	− 0.11	0.48	.83	0.90	0.35,2.32	0.26	0.12	**.029**	1.30	1.03,1.65
Latency	− 0.05	0.36	.90	0.96	0.47,1.93	0.29	0.10	**.005**	1.34	1.09,1.64
Duration	− 0.60	0.37	.10	0.55	0.26,1.13	0.11	0.10	.29	1.12	0.91,1.36
Efficiency	− 0.30	0.34	.38	0.74	0.38,1.45	0.03	0.10	.76	1.03	0.85,1.25
Disturbances	0.66	0.55	.23	1.93	0.66,5.68	0.47	0.14	**.001**	1.60	1.21,2.12
Medication	0.08	0.42	.86	1.08	0.47,2.47	0.30	0.11	**.007**	1.35	1.09,1.69
Daily Dys.	− 0.02	0.32	.95	0.98	0.52,1.84	0.32	0.10	**.002**	1.38	1.12,1.69

Abbreviations: CI, confidence interval; Daily Dys., daily dysfunction; OR, odds ratio; SE, standard error.
p-values < .05 are bolded.

we cannot rule out the possibility that the observed association between prior concussion and sleep quality is due to individuals with poor sleep being predisposed for increased risk of concussion. Certainly, additional research is needed to determine the complex interrelationship of concussion, sleep, and adverse health outcomes observed in some athletes later in life.[2,34]

Current results also extend upon the current literature by investigating the association of concussion and RHI history with specific subcomponents of self-reported sleep. In the primary analyses, concussion history was associated with worse overall sleep quality, greater sleep latency, more sleep disturbances or waking, increased use of sleep aids or medication, and increased reports of daily dysfunction. In contrast, there was no association of concussion history with self-reported total hours of sleep or sleep efficiency (i.e., the ratio of hours spent in bed vs. hours asleep), even though participants with more concussions felt they take longer to fall asleep and wake up more throughout the night. This pattern of findings is consistent with factor analyses of PSQI subcomponent scores that have found that sleep efficiency and sleep duration generally load together (e.g., a "sleep efficiency" factor), while the other subcomponents load together as either one or two additional factors.[35-37] Thus, current results suggest that cumulative concussions are associated with perceived sleep quality and disturbances. From a research perspective, these results demonstrate that questions regarding how many hours a participant slept last night are not sufficient when attempting to determine the sleep-related sequelae of cumulative concussion. Moreover, these results also highlight the need for future studies aimed at determining if specific components of sleep quality are more predictive of adverse long-term outcomes observed in some former athletes with extensive concussion history,[34] and if so, identifying candidate interventions to potentially improve sleep quality in these individuals (e.g., limiting disturbances at night). It should be noted, however, that PSQI subcomponents scores in the current study were binarized for analysis based on the distribution of collected responses. While this is a parsimonious approach, we do not suggest that these are true cutoffs to reliably reflect "impaired" and "nonimpaired" groups, but rather this should be interpreted as a two-level variable that reflects severity of the measured sleep construct in this cohort.

Current findings also demonstrate that the observed associations of prior concussion and sleep were largely consistent across multiple operational definitions of concussion that differed based on their reliance on symptoms alone (in conjunction with injury mechanisms) vs. the presence of alterations in mental status or consciousness. We have previously found that using the more conservative definition requiring alteration in mental status (i.e., presence of symptoms alone is not sufficient for diagnosis) results in stronger associations of prior concussion with symptoms as well as hippocampal volumes.[24] Similarly, acutely, injuries with either LOC or PTA have been shown to result in greater elevation in blood-based TBI biomarkers and hippocampal volumes, suggesting more severe physiological injury.[27,28,38] In the current study, it is noteworthy that qualitatively, higher ORs were generally observed for the ACRM definition (i.e., requiring alteration in mental status) compared to the CISG definition (i.e., symptoms alone sufficient). ORs were even higher in models investigating number of prior concussions with either LOC or PTA, though they were only significantly associated with sleep disturbances and daily dysfunction, likely due to the fewer number of total injuries meeting this set of criteria. Regardless, current results highlight the importance of efforts aimed at reducing concussion risk in athletes, such as changes to preseason practice guidelines, age-appropriate rule changes, and coaching aimed at developing safe techniques for tackling and related high-risk activities.[39]

In contrast to our findings on concussion history, this study found no evidence for a relationship between RHI and sleep quality, regardless of how RHI was operationalized (e.g., AFE vs. years of exposure). Younger AFE was recently associated with greater sleep disturbances in former NFL players,[17] while RHI was identified as a risk factor for REM sleep disorders in CTE.[18] It is possible that RHI exposure effects do not manifest until later in life or only emerge with aging in conjunction with various lifestyle or other moderating factors, and thus were not observable in the current younger cohort. Longitudinal studies are needed to determine the trajectory of sleep quality as it relates to RHI exposure in active and former athletes.

Strengths and limitations

The current results should be interpreted within the contexts of the strengths and limitations of the study. First, the current study had strict exclusion criteria. While this may limit generalizability of findings to some extent, it also limits the possibility that current associations are observed by underlying comorbidities (e.g., mood disorders). Second, this study included male and female athletes from a variety of sports, with varying levels of RHI exposure and concussion history. Both RHI and concussion history were determined retrospectively and are potentially impacted by recall bias. Semistructured interviews, however, allowed retrospective quantification of concussion and RHI history based on different operational definitions. This, in theory, should provide accurate estimates of both measures, while also allowing the effects of RHI and concussion on sleep to be modeled separately. One limitation of the current study is that the study population is relatively homogenous, and future studies in more diverse (i.e., racial, ethnic, and age) cohorts are needed to determine the generalizability of current fundings. In addition, sleep quality was assessed based on self-report using the PSQI. The PSQI has been shown to be reliable and valid with high convergence with sleep diary data,[37,40] but it has lower correspondence to lab-based polysomnography data.[41] Ultimately, future studies assessing relationships of cumulative concussion with more objective measures of sleep quality are needed. Finally, secondary and exploratory analyses were purposely conducted without strict correction for multiple comparison correction due to their complementary nature, reducing the likelihood of type II error at the risk for type I error.

Conclusions

Healthy, collegiate-aged athletes with more prior concussions report worse sleep in a dose-dependent manner. Investigation of specific subcomponents of sleep suggests that concussion history is associated with multiple aspects of subjective sleep quality rather than sleep quantity. Sleep represents an important factor to consider for future research aimed at characterizing and ultimately preventing adverse long-term health outcomes that have been reported in current and former athletes with repeated concussion.

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Declaration of conflicts of interest

Dr Meier receives compensation as a member of the Clinical and Scientific Advisory Board for Quadrant Biosciences Inc. All other authors report no disclosures.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.sleh.2024.04.005.

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